AP CALCULUS AB

Unit 1 • Practice

# Limits and Continuity

AP-style practice • Multiple choice and free response

**About this practice set**

The questions follow the format of the AP Calculus AB exam. Part A and Part B are multiple choice; Part C is free response. Each question is labeled **NO CALCULATOR** or **CALCULATOR**. Suggested timing: Part A 40 min • Part B 8 min • Part C 45 min.

## Part A • Multiple choice • No calculator

**Directions • Part A**

A calculator may **not** be used on this part. Solve each problem, then choose the best of the four answer choices. Unless otherwise stated, the domain of a function $f$ is the set of all real numbers $x$ for which $f(x)$ is a real number. Some questions refer to a graph: read the values you need from the grid.

### Question 1 (1.6)

NO CALCULATOR

$$\lim_{x \to 3} \frac{x^2 - 9}{x^2 - 2x - 3} \text{ is}$$

(A) $0$

(B) $\frac{3}{2}$

(C) $3$

(D) nonexistent

### Question 2 (1.6)

NO CALCULATOR

$$\lim_{x \to 0} \frac{\sqrt{x+9} - 3}{x} \text{ is}$$

(A) $0$

(B) $\frac{1}{3}$

(C) $\frac{1}{6}$

(D) $3$

### Question 3 (1.8)

NO CALCULATOR

$$\lim_{x \to 0} \frac{\sin(5x)}{x} \text{ is}$$

(A) $5$

(B) $1$

(C) $\frac{1}{5}$

(D) $0$

**Question 4** (1.15) NO CALCULATOR

$\lim_{x\to\infty} \dfrac{4x^3 - x}{2x^3 + 7x^2}$ is

(A) 0
(B) $\dfrac{4}{7}$
(C) $\infty$
(D) 2

**Question 5** (1.15) NO CALCULATOR

$\lim_{x\to-\infty} \dfrac{2x}{\sqrt{x^2+5}}$ is

(A) 2
(B) $-2$
(C) 0
(D) $-\infty$

**Question 6** (1.7) NO CALCULATOR

$\lim_{x\to2^-} \dfrac{|x-2|}{x-2}$ is

(A) 1
(B) 0
(C) $-1$
(D) nonexistent

**Question 7** (1.4) NO CALCULATOR

| $x$ | 0.9 | 0.99 | 0.999 | 1.001 | 1.01 | 1.1 |
|---|---|---|---|---|---|---|
| $f(x)$ | 2.71 | 2.97 | 2.997 | 3.003 | 3.03 | 3.31 |

Based on the table, which is the best estimate of $\lim_{x\to1} f(x)$?

(A) 2.997
(B) 3
(C) 3.003
(D) The limit does not exist.

**Question 8** (1.11) NO CALCULATOR

Let $f(x) = \begin{cases} x^2+1, & x<1 \\ 3x-1, & x\geq 1. \end{cases}$ Which statement is true?

(A) $\lim_{x\to1} f(x)$ does not exist.
(B) $\lim_{x\to1} f(x)$ exists, but $f$ is not continuous at $x=1$.
(C) $f$ is continuous at $x=1$.
(D) $f(1)$ is undefined.

**Question 9** (1.13) NO CALCULATOR

For what value of $k$ is $f(x) = \begin{cases} kx+3, & x \le 2 \\ x^2 - k, & x > 2 \end{cases}$ continuous at $x = 2$?

(A) $-\frac{1}{3}$
(B) $\frac{1}{3}$
(C) $1$
(D) $7$

**Question 10** (1.10) NO CALCULATOR

What type of discontinuity does $g(x) = \frac{x^2-4}{x-2}$ have at $x = 2$?

(A) Jump
(B) Infinite
(C) Removable
(D) None; $g$ is continuous at $x = 2$.

**Question 11** (1.14) NO CALCULATOR

Which line or lines are vertical asymptotes of the graph of $h(x) = \frac{x+1}{x^2-1}$?

(A) $x = 1$ only
(B) $x = -1$ only
(C) $x = 1$ and $x = -1$
(D) $y = 0$ only

**Question 12** (1.8) NO CALCULATOR

$\lim_{x \to 0} x^2 \cos\left(\frac{1}{x}\right)$ is

(A) $1$
(B) $0$
(C) $-1$
(D) nonexistent

**Question 13** (1.16) NO CALCULATOR

The function $f$ is continuous on $[1,5]$ with $f(1) = -2$ and $f(5) = 6$. Which statement must be true?

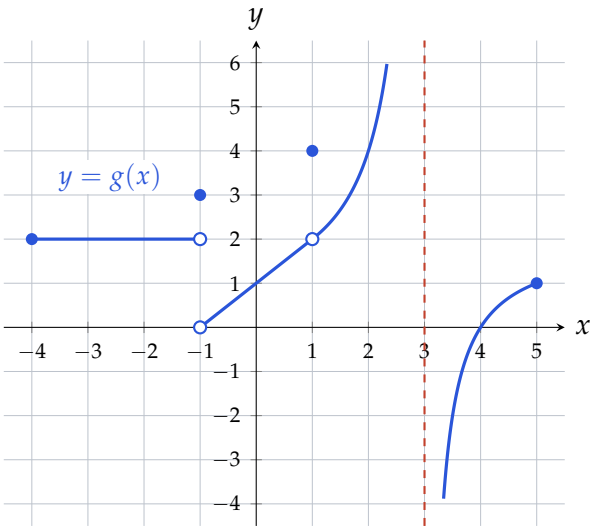
(A) $f(c) = 8$ for some $c$ in $(1,5)$.
(B) $f(3) = 2$.
(C) $f$ is increasing on $[1,5]$.
(D) $f(c) = 4$ for some $c$ in $(1,5)$.

**Question 14** (1.15) NO CALCULATOR

$\lim_{x \to \infty} \frac{3e^x + x^2}{e^x - 1}$ is

(A) 0
(B) 1
(C) 3
(D) $\infty$

**Questions 15 to 19 refer to the graph of the function $g$ below.**




The domain of $g$ is $[-4,3) \cup (3,5]$. The line $x = 3$ is a vertical asymptote.

**Question 15** (1.3 • graph) NO CALCULATOR

Which of the following limits does **not** exist?

(A) $\lim_{x \to 1} g(x)$
(B) $\lim_{x \to -1^+} g(x)$
(C) $\lim_{x \to -1} g(x)$
(D) $\lim_{x \to 4} g(x)$

**Question 16** (1.3 • graph) NO CALCULATOR

What is $\lim_{x \to 1} g(x)$?

(A) 4
(B) 2
(C) 1
(D) It does not exist.

**Question 17** (1.14 • graph) NO CALCULATOR

Which statement about the behavior of $g$ near $x = 3$ is true?

(A) $\lim_{x \to 3^-} g(x) = +\infty$ and $\lim_{x \to 3^+} g(x) = -\infty$
(B) $\lim_{x \to 3^-} g(x) = -\infty$ and $\lim_{x \to 3^+} g(x) = +\infty$
(C) $\lim_{x \to 3} g(x) = +\infty$
(D) $\lim_{x \to 3} g(x) = 2$

**Question 18** (1.10 • graph) NO CALCULATOR

At which value of $x$ does $g$ have a removable discontinuity?

(A) $x = -1$
(B) $x = 3$
(C) $x = -4$
(D) $x = 1$

**Question 19** (1.9 • graph) NO CALCULATOR

What is $\lim_{x \to 0} g\big(g(x)\big)$?

(A) 1
(B) 4
(C) 2
(D) It does not exist.

## Part B • Multiple choice • Graphing calculator

**Directions • Part B**

A graphing calculator is **required** for the questions in this part. If the exact value of the correct answer is not one of the choices, choose the one that best approximates it.

**Question 20** (1.4) CALCULATOR

Using a table of values or a graph, what is $\lim_{x \to 0} \dfrac{2^x - 1}{x}$, to three decimal places?

(A) 0.500
(B) 0.693
(C) 1.000
(D) 1.443

**Question 21** (1.16) CALCULATOR

How many values of $c$ in the open interval $(0, 2)$ satisfy $c^3 - 4c + 1 = 0$?

(A) None
(B) One
(C) Two
(D) Three

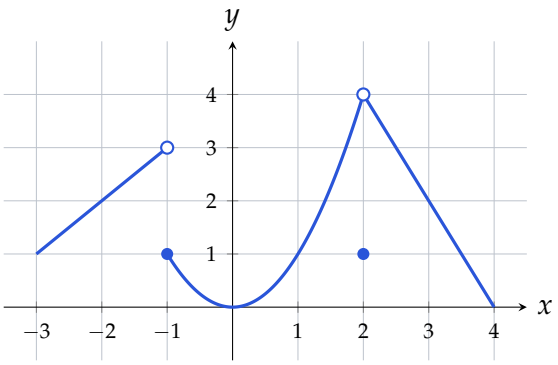
## Part C • Free response

**Directions • Part C**

Show all of your work: your answers are graded on the correctness and completeness of your methods as well as on the final answers. Justify answers with mathematical reasons, naming the theorem or definition you use. Use a calculator only on the question labeled CALCULATOR. Decimal answers should be correct to three places after the decimal point.

**Question 22** (Graph analysis) NO CALCULATOR

The graph of the function $f$ on $[-3, 4]$ is shown. It consists of a line segment for $-3 \le x < -1$, the parabola $y = x^2$ for $-1 \le x < 2$, and a line segment for $2 < x \le 4$. Also $f(-1) = 1$ and $f(2) = 1$.




(a) Find $\lim_{x \to -1^-} f(x)$ and $\lim_{x \to -1^+} f(x)$. Does $\lim_{x \to -1} f(x)$ exist? Give a reason for your answer.
(b) Find $\lim_{x \to 2} f(x)$. Is $f$ continuous at $x = 2$? Name the type of discontinuity, if any, and explain how $f(2)$ could be redefined to make $f$ continuous at $x = 2$.
(c) Find $\lim_{x \to 3} f(f(x))$. Show the work that leads to your answer.
(d) $f(1) = 1$ and $f(3) = 2$. Explain why the Intermediate Value Theorem cannot be used to guarantee a value $c$ in $[1, 3]$ with $f(c) = 1.5$. Does such a value exist? Justify.

**Question 23** (Algebraic limits and continuity) NO CALCULATOR

Let $g(x) = \begin{cases} \dfrac{x^2 + x - 6}{x - 2}, & x < 2 \\ cx + 1, & x \ge 2 \end{cases}$ where $c$ is a constant, and let $h(x) = \dfrac{x^2 + x - 6}{x^2 - 4}$.

(a) Find $\lim_{x \to 2^-} g(x)$. Show the work that leads to your answer.
(b) Find the value of $c$ for which $g$ is continuous at $x = 2$. Justify your answer using the definition of continuity.
(c) Find all values of $x$ where $h$ is not continuous, and classify each discontinuity. Justify with limits.
(d) Find $\lim_{x \to \infty} h(x)$ and $\lim_{x \to -2^+} h(x)$.

**Question 24** (Table and theorems) CALCULATOR

The temperature $W$, in degrees Celsius, of a greenhouse is a continuous function of time $t$, in hours, for $0 \le t \le 10$. Selected values are given in the table.

| $t$ (hours) | 0 | 3 | 5 | 8 | 10 |
|---|---|---|---|---|---|
| $W(t)$ (°C) | 20 | 24 | 31 | 27 | 22 |

(a) Must there be a time $t$ in $[0, 10]$ at which $W(t) = 30$? Justify your answer.
(b) Explain why there must be at least two times in $[0, 10]$ at which $W(t) = 25$.
(c) Use the data to estimate the rate at which the temperature is changing at $t = 4$ hours. Show the computation and give units.
(d) A function $f$ satisfies $4x - x^2 \le f(x) \le x^2 - 4x + 8$ for all $x$. Find $\lim_{x \to 2} f(x)$, and name the theorem you use.

Check your answers with the answer key for this unit.